
Appendix B - Test report maintenance

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- Print the following pages of the appendix B.
- Description of every single point of the report see chapter 5.

* Please, always pay attention to the updated service info relating to a more recent version.



Caution

The maintenance procedures must be performed once a year or after every 10.000 sealing processes.



Note

Maintenance procedures, adjustments, modifications or repairs may only be carried out by the manufacturer or persons authorized by him.

Manufacturer:	Date of installation:
Device:	Technician:
Software Version:	Place:
Serial number:	Service report no.:
Run Counter:	
E-Code:	Language:

No.	Description	Function	Meas. value	Checked	Repaired	n.A
1 Exterior of the device						
1.1	Cleaning	Thoroughly clean the device, especially clamps and electrodes				
1.2	Rear panel	Check the strain relief of the power cable				
		Labels, labeling				
		Damage, cracks				
1.3	Keypad / display	Damage				
		Function				
		Action point				
1.4	Tubing cover switch	Orientation, switching point				
1.5	Heating element	Damage				
2 Interior of the device						
2.1	Cleaning					
2.2	Threaded spindle, as required	Silicone oil (part no.: M642401)				
2.3	Guides	Silicone oil (part no.: M642401)				

No.	Description	Function	Meas. value	Checked	Repaired	n.A
3	Parameter values					
●	Reading out parameters: Before performing maintenance measures or replacing the P.C.B., read out all parameter values from the service menu and record them in the maintenance report or make a back-up with the corresponding Service tool: After replacing the P.C.B., readjust or upload the following parameter values in the service menu. Then perform a visual inspection/functional check. – Xmotor Outside pos – Xmotor pos.dock – Heater dock temp. – Xmotor heat pos – heating Parameter/X-MotorSpeed Heat. If it is not possible to read out the parameter values from the service menu, the above-mentioned values must be determined or re-adjusted. The default setting will be used for all other parameter values. Then perform a visual inspection/functional check.					
3.1	Voltage Supply 5 Volt ≥ E-Code 140	4.60 V - 5.40 V	V			
		4.8 V - 5.1 V	V			
3.2	Voltage Supply 24 Volt	23.65 V - 24.31 V	V			
3.3	Display brightness (only P.C.B. LP1058-1) ≥ E-Code 140	5.5 ±0.1 V	V			
		95 mV - 105 mV	mV			
3.4	Blower current ≥ E-Code 140	8.6 mA - 20.9 mA	mA			
		28.6 mA - 98.1 mA	mA			
3.5	Xmotor Outside pos < E-code 030 ≥ E-code 030 or ≥ software 2.2x	0.2 s - 1.5 s	s			
		95 counts	counts			
		100 counts	counts			
3.6	Y motor	0.8 s - 2.5 s	s			
3.7	HF motor	0.5 s - 1.7 s	s			
3.8	Dock cooldown time	4 s	s			
3.9	Xmotor pos.dock	Variable	counts			
3.10	Heater dock temp.	typical value 120 counts 465 °C ± 5 °C (China version: 470 °C ±5°C)	counts °C			

No.	Description	Function	Meas. value	Checked	Repaired	n.A
3.11	Tube heating time	12 s	s			
3.12	Xmotor heat pos	Variable	s			
3.13	Seal stretch Tubes	3 - 45 counts	counts			

No.	Description	Function	Meas. value	Checked	Repaired	n.A
4 P.C.B. LP 1058 microprocessor board						
4.1	Unregulated voltage (only E-code <069)	(31–35.5 V)	V			
4.2	Display contrast	Adjust with P2 / P.C.B. LP 1058				
5 HF electrodes unit						
5.1	HF motor	Check the toothed belt				
5.2	HF electrodes	Damage Cleaning				
5.3	HF-0 position switch The HF electrodes must be open so far that a tubing may be easily inserted	5 mm ± 0.5 mm	mm			
5.4	HF-1 position switch Both HF electrode pairs must touch each other simultaneously.					
6 Tube clamps						
6.1	Docking position alignment In the docking position (X3/Y1) the closed clamping jaws are in one line					
6.2	Height position alignment In initial position (X0/Y0), the adjusting gauge is positioned evenly on the tube clamps and on the central block. Depending on the E-code					
6.3	Alignment with the electrodes In initial position (X0/Y0), the closed electrodes are in one line with the closed clamping jaws.					
6.4	Alignment with the central block In initial position (Y0/X0), with open tube clamps and open electrodes, an inserted 4.2 mm tubing can easily be moved.					

No.	Description	Function	Meas. value	Checked	Repaired	n.A
7 X system						
7.1	X motor	Check the toothed belt				
7.2	Tube clamps docking position Determine the distance of the clamps in docking position using a feeler gauge.	(0.6 mm ±0.05 mm) (China version: 0.65 mm ±0.05 mm) Fan on x-motor pos counts (act)	mm			
			counts			
8 Y system						
8.1	Y motor	Check the toothed belt				
8.2	Y0 position switch					
8.3	Y1 position switch					
9 Heating element						
9.1	H0 position switch The switching lug just must have some play at the end stop.					
9.2	H1 position switch The switching lug just must have some play at the end stop.					
9.3	Heating position of the tube clamps Determine the distance between the two tube clamps and the central block Distance between the tubing ends and the heating element Adjustment of the maximum heating element temperature	(min. 2 mm) (0.8 mm ±0.05 mm) Temperature (465 °C ±5 °C) (China version: 470 °C ± 5 °C) Temperature value (counts)	mm			
			mm			
			°C			
			counts			

No.	Description	Function	Meas. value	Checked	Repaired	n.A
10 HF voltage						
10.1	With empty tubing					
	Initial voltage values of the left electrode	UL_ini.	V			
	Initial voltage values of the right electrode	UR_ini.	V			
	Minimum peak voltage of the smaller sealing curve	(4.1 V $\pm$ 0.1 V)	V			
10.2	With tubing filled with NaCl					
	Initial voltage values of the left electrode	UL_ini.	V			
	Initial voltage values of the right electrode	UR_ini.	V			
	Difference between the initial values for the filled tubing and the empty tubing	must range between - 0.1 V and + 0.3 V	V			
11 Seal test						
11.1	Visual inspection of the sealing seams					

No.	Description	Function	Meas. value	Checked	Repaired	n.A
12	Dock test					
12.1	Tightness of the sealing seams	Perform several sealing processes				
	(L) Fresenius 4.2 dry with (R) Fresenius 4.2 dry	M600241 M600241				
	(L) Fresenius 4.2 filled with (R) Fresenius 4.2 filled	M600241 + NaCl M600241 + NaCl				
	Fresenius 4.2 filled with Fresenius 4.2 dry	M600241 + NaCl M600241				
	Customer tube Part number Manufacturer of tube Specification	Combination docked wet + wet wet + dry dry + dry				

No.	Description	Function	Meas. value	Checked	Repaired	n.A
13 Electrical test according to DIN VDE 0701-0702						
13.1	Equivalent leakage current: (L1 and N against PE)	<3.5 mA (230 V) <0.7 mA (110/120 V)	mA			
13.2	Check the following test points using the probe tip: – 6 x screws for housing rear panel – 2 x screws for housing front panel – 2 x Allen screw, central block – The value of 0.2 Ω must not be exceeded.	Nominal value: < 300 mΩ	mΩ			
13.3	Resistance (L1 and N against PE)	Nominal value: > 2 MΩ	MΩ			
14 Overview of the used calibration tools						
14.1	Test location / inventory number of the tool					

On completion of the maintenance procedures, start the **Life test** service menu and have it run for approx. 5 minutes.

Access the Runcounter service menu and reset the counter reading.

I herewith confirm proper performance of any maintenance work as specified above, and the data recorded to be true and correct.			
Technician's signature: 	Date: 	Customer's signature: 	Date:

Customer report sheet

Data of customer/device:

Manufacturer:	Date of installation:
Device:	Technician:
Software Version:	Place:
Serial number:	Service report no.:
Run Counter:	
E-Code:	Language:

No.	Description	Checked	Repaired
01	Exterior of the device		
02	Interior of the device		
03	Parameter values		
04	P.C.B. LP 1058 microprocessor board		
05	HF electrodes unit		
06	Tube clamps		
07	X system		
08	Y system		
09	Heating element		
10	HF voltage		
11	Seal test		
12	Dock test		
13	Electrical test according to DIN VDE 0701-0702		
14	Overview of the used calibration tools		

I herewith confirm proper performance of any maintenance work as specified above, and the data recorded to be true and correct.			
Technician's signature:	Date:	Customer's signature:	Date:
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History

Index	Modification	Date	Name
00	Initial issue	17.07.2020	O. Dakni
01	No. 14 added	01.06.2021	

